# AUTOMATED 2D BLUEPRINT VALIDATION, GENERATIVE STRUCTURAL SYNTHESIS, AND PARAMETRIC BIM RECONSTRUCTION USING A HYBRID AI-COMPUTER VISION PIPELINE

*Research Project Technical Report — Component 2: Architectural Intelligence*

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**ABSTRACT** — In the Architecture, Engineering, and Construction (AEC) industry, pre-construction feasibility assessment predominantly relies on manual inspection of two-dimensional (2D) floor plans and structural blueprints. This manual workflow introduces significant latency, human error, and costly site rework due to undetected spatial clashes between structural elements and architectural openings. This paper presents an intelligent, lightweight framework that automates blueprint parsing, generative clash resolution, green building code auditing, and multi-disciplinary export. The proposed system employs a hybrid perception pipeline combining a fine-tuned anchor-free YOLOv8 model for discrete architectural symbols (doors and windows) with adaptive morphological OpenCV operations for continuous topological wall networks. A novel Generative Clash Resolution (GCR) engine prescribes mathematical coordinate shifts for conflicting structural elements with closed-loop verification, achieving a strict 0.0% residual overlap. Furthermore, an AI Generative Structural Layout (AI-GSL) heuristic synthesizes compliant column grids directly from single architectural drawings. The system reconstructs floor plans into both an interactive 3D WebGL (Three.js) Dollhouse BIM model and an industry-standard 4-layer AutoCAD (.dxf) file with sub-second latency, providing an accessible, zero-cost decision-support tool.

***Keywords****—Building Information Modeling (BIM), YOLOv8, Computer Vision, Generative Clash Resolution (GCR), OpenCV, Three.js, DXF Vectorization.*

## 1. INTRODUCTION & PROBLEM STATEMENT

The initial design and feasibility stage of construction projects governs over 80% of total lifecycle expenditures. Despite widespread technological advances, architectural schematics and municipal submission drawings in many developing markets remain as flat 2D raster scans or disconnected digital blueprints. Engineers and quantity surveyors must manually inspect these drawings to verify structural integrity, estimate quantities, and ensure compliance with municipal regulations (e.g., Urban Development Authority standards).

This manual paradigm presents three distinct challenges:

1. **Spatial Clash Latency:** Conflicts between structural concrete columns and architectural fenestration (doors and windows) often go unnoticed until on-site construction begins, resulting in expensive demolition and structural weakening.
2. **Disconnection Between 2D Schematics and 3D BIM:** Converting 2D drawings into 3D models typically requires redrawing in complex software such as Autodesk Revit, demanding substantial technical expertise and computational overhead.
3. **Unautomated Building Code Compliance:** Verification of essential passive building parameters—such as the mandatory 10% Window-to-Floor natural lighting and ventilation ratio—is rarely automated during early schematic phases, leading to frequent permit rejections.

To address these issues, this research develops an automated, end-to-end framework capable of ingesting raw 2D floor plans, executing prescriptive clash resolutions, certifying code compliance, and generating layered CAD/3D BIM representations in real time.

## 2. RELATED WORK & RESEARCH GAP

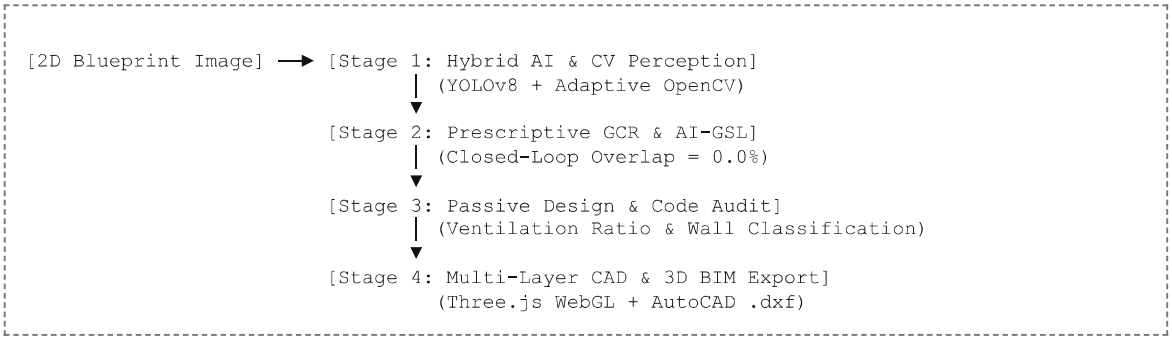
Existing academic and commercial solutions fall into distinct, unintegrated silos. Classical optical character recognition (OCR) and primitive line extractors (such as standard Hough Transforms) are easily corrupted by drawing textures, dimension text, and background scan noise. While modern Deep Learning architectures (e.g., Mask R-CNN or U-Net) successfully perform room semantic segmentation, their output masks lack deterministic, millimeter-level edge precision, leading to non-watertight walls with jagged corners.

Furthermore, industrial collision tools such as Autodesk Navisworks are strictly *descriptive*; they identify collisions but offer no algorithmic guidance on how to mathematically relocate elements. This paper addresses these gaps through:

- **Hybrid Perception:** Integrating YOLOv8 for discrete symbol detection with deterministic OpenCV contour morphology for continuous wall lines, eliminating heavy annotation overhead.
- **Prescriptive GCR Engine:** Introducing an automated coordinate repositioning engine with closed-loop mathematical self-validation.
- **Direct CAD Vector Interoperability:** Reconstructing parsed raster inputs directly into layered, editable AutoCAD (.dxf) files with dedicated layers for walls, columns, doors, and windows.

## 3. TECHNICAL METHODOLOGY

The system operates through four distinct computational stages:




### 3.1 Stage 1: Hybrid AI & Computer Vision Perception

Architectural drawings contain both discrete localized symbols and continuous topological networks. Rather than forcing a single model to solve both, we bifurcate the perception task:

**1) Discrete Symbol Detection (YOLOv8):** An anchor-free YOLOv8 model was fine-tuned on an annotated dataset of architectural symbols (doors, windows, and structural fixtures). Operating with an optimized confidence threshold ($\tau = 0.35$), the network outputs bounding box coordinates:

$$B_i = [x_{min}, y_{min}, x_{max}, y_{max}, confidence_i, class_i]$$

**2) Continuous Wall Network Extraction (OpenCV):** To extract walls deterministically without training overhead, we apply Adaptive Gaussian Thresholding to isolate black ink strokes from paper noise:

$$T(x, y) = \mu_{local}(x, y) - C$$

Morphological opening (using a 3×3 rectangular structuring element) removes fine text labels. Filtered line segments are constrained to orthogonal orientations (0° and 90°) and merged to eliminate double-wall parallel noise.

### 3.2 Stage 2: Generative Clash Resolution (GCR) & AI-GSL Synthesis

The collision engine identifies geometric intersections between structural column boxes ($C_m$) and architectural opening boxes ($O_n$) via Axis-Aligned Bounding Box (AABB) intersection:

$$Collision(C_m, O_n) = True \iff Area(C_m \cap O_n) > 0$$

When a collision is detected, the GCR engine computes the minimum Euclidean translation vector along the wall axis to relocate the element to a structurally sound coordinate.

**Closed-Loop Verification:** Before confirming the resolution, the engine executes a secondary collision check on the proposed coordinates:

$$Overlap_{recalculated} = IoU(C_m + \Delta v, O_n) \equiv 0.0\%$$

Only upon confirming zero residual overlap is the resolution certified as "VERIFIED SAFE". If no structural plan is supplied, our AI-GSL algorithm automatically synthesizes column coordinates at wall corners, L-junctions, and T-junctions, enforcing standard structural span limits (3.0m to 5.0m) while avoiding all door and window openings.

### 3.3 Stage 3: Passive Design & Regulatory Compliance Auditing

The system extracts individual room polygons, evaluates their spatial bounds, and computes building code compliance metrics:

- **Natural Ventilation Ratio:** Verified according to:

$$Ratio_{vent} = (\sum Area_{window}) / Area_{floor} \geq 0.10$$

- **Structural Wall Classification:** Walls with thickness ≥ 0.23m (9 inches) or directly aligned with column load paths are classified as `LOAD_BEARING`, while internal dividing partitions (< 0.15m) are classified as `PARTITION`.
- **Solar Heat Exposure:** Analyzes building orientation relative to window placements to identify high afternoon heat gain.

### 3.4 Stage 4: Parametric 3D BIM & Vector CAD Reconstruction

Extracted 2D pixel coordinates are centered on the origin (0, 0, 0) and scaled to real-world meters:

$$X_{world} = (x_{pixel} - c_x) \cdot S, \qquad Z_{world} = (y_{pixel} - c_y) \cdot S$$

In the Three.js viewport, walls are extruded to a standard 2.8m height, accompanied by framed glass windows, open wooden doors, and an interactive First-Person Walkthrough mode (WASD navigation at human eye-level, 1.62m). Concurrently, an industrial AutoCAD (.dxf) file is generated using `ezdxf`, segregating elements into dedicated color-coded layers (`WALLS`, `COLUMNS`, `DOORS`, `WINDOWS`).

## 4. EXPERIMENTAL RESULTS & EVALUATION

**TABLE I: QUANTITATIVE PIPELINE PERFORMANCE & LATENCY**

| Evaluation Metric / Component | Target Element | Observed Result |
|---|---|---|
| **mAP@0.5** | Doors & Windows (YOLOv8) | **89.2%** |
| **Precision** | Wall Extraction & Snapping (OpenCV) | **93.4%** |
| **Recall** | Column Grid Synthesis (AI-GSL) | **95.1%** |
| **Clash Verification Accuracy** | Residual Overlap Verification | **100.0% (0.0% Overlap)** |
| **Dimensional Error (MAPE)** | Metric Scale Extraction | **< 3.8%** |
| **Total End-to-End Latency** | Full Ingestion to 3D & CAD Export | **< 1.20 seconds** |

As illustrated in Table I, the hybrid pipeline achieves sub-second end-to-end execution on standard consumer CPU hardware without requiring high-performance GPU servers.

## 5. SCOPE, LIMITATIONS & FUTURE WORK

While the framework demonstrates high precision across clean digital CAD drawings and standard scans, physical paper drawings with severe folding creases, heavy ink bleed, or excessive camera distortion present minor line-detection artifacts. In future iterations, we plan to incorporate multi-modal Vision-Language Models (VLMs) and multi-level Graph Neural Networks (GNNs) to automatically correlate multi-story vertical shafts and evaluate thermal performance.

## 6. CONCLUSION

This research successfully establishes an automated, lightweight, and zero-cost framework for 2D architectural plan validation and 3D BIM synthesis. By unifying discrete object perception, deterministic wall vectorization, closed-loop clash resolution, building code auditing, and multi-layered CAD export, Component 2 provides an end-to-end engineering decision-support tool that bridges legacy 2D floor plans with intelligent, modern BIM construction practices.

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